

GCC Port Efficiency & Diversion Readiness

Decision support for regional port monitoring and disruption planning

Business need

A regional importer relies on major GCC ports to maintain dependable supply routes. When port performance weakens or a primary gateway is disrupted, planners need a consistent way to identify the change, compare the affected port with relevant peers, and decide which alternatives should be investigated first. The available CPPI data is comprehensive, but it does not provide a focused GCC decision view without further modelling and analysis.

Project objective

Build an interactive Power BI report using World Bank CPPI data for 2020-2025. The report should monitor regional performance, provide selected port diagnostics, and screen data for eligible alternatives during a disruption scenario. It is intended to support early investigation and planning; it is not a substitute for live operational or routing analysis.

Decisions to support

- Is average GCC port performance improving or deteriorating over time?
- Which ports require attention in the selected year, and which ports lead the regional ranking?
- How does a selected port compare with other ports in the same maritime services region?
- Which data eligible ports should be reviewed first if a selected gateway is disrupted?

Primary users

- Supply chain and logistics managers responsible for regional performance and continuity planning.
- Operations and planning analysts responsible for investigating port changes and reviewing alternatives.

- BI and reporting teams responsible for maintaining the model, measures, and data refresh process.

SCOPE AND REQUIREMENTS

The report must make performance changes visible and recommendations explainable

Project scope

In scope: Bahrain, Kuwait, Oman, Qatar, Saudi Arabia and the United Arab Emirates; annual CPPI performance from 2020-2025; GCC trends and ranking; selected port peer analysis; volatility; disruption screening; and port level drillthrough.

Out of scope: Live spare capacity, shipping schedules, sailing distance, cargo compatibility, customs constraints, inland transport, landed cost and causal analysis of port performance.

Data and modelling approach

The source is the World Bank CPPI 2025 Annex, containing 426 port records and 20 fields. The GCC view covers 13 ports with a reported 2025 score. Annual score columns are reshaped into a port year fact table, supported by reusable port and year dimensions. The 2025 diagnostic fields remain at port grain, while disconnected selector tables support peer and disruption scenarios without creating ambiguous relationships.

Core requirements

- A year selector must update regional KPIs, deterioration counts, ranking and comparison visuals.
- A port selector must update the annual trend, latest KPIs and regional peer comparison.
- The regional peer average must exclude the selected port from its own benchmark.
- The disruption selector must remove the disrupted port while retaining the alternative candidate population.
- The readiness screen must weight latest CPPI at 50%, historical stability at 30% and recent trend at 20%.
- Drillthrough must expose the annual history and measures supporting a recommendation.

DELIVERY AND ACCEPTANCE

The solution is complete when it is accurate, usable and ready to hand over

Expected deliverables

- Power BI report with Executive Overview, Trends & Peer Analysis and Diversion Scenario pages, plus a hidden Port Drillthrough page.
- Reusable semantic model, Power Query transformations and documented DAX measures.
- Business brief, case study, dashboard PDF, screenshots and GitHub README.

Acceptance criteria

- The 2025 GCC average reconciles to 55.98 across the 13 ports with a reported current score.
- Selected port KPIs reconcile to the source and respond correctly to filter context.
- For a Jebel Ali disruption, 11 alternatives qualify and Salalah ranks first with a readiness score of 89.0.
- Sorting, conditional colors, drillthrough context and back-button navigation work consistently.

Assumptions and constraints

The GCC average is an unweighted port level mean, and historical calculations use available observations. CPPI measures port efficiency using vessel time in port, it does not explain the cause of a score or confirm operational feasibility. The readiness output is therefore a transparent performance screen rather than a final diversion decision.

Data attribution

World Bank CPPI 2025 overview and dataset

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